2026 Kentucky Cybersecurity and Forensics Conference

STUDENT POSTER ABSTRACTS

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Harold R. Benson Research and Demonstration Farm

Building Cyber Resilience Through Collaboration: Academia, Industry, and Government

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Student Poster Session

Time: 12:00 pm – 1:00 pm

JUDGES

Elwyn Rainer II

James Meece

Sheri Donahu

UAV-Based Multispectral Remote Sensing and Machine Learning for Soybean Yield Prediction

Susma Adhikari
Graduate Student and Research Assistant
Kentucky State University
Session: Student Poster Session | **12:00 pm – 1:00 pm**

ABSTRACT

Accurate in-season soybean yield prediction can improve precision agriculture and nitrogen management. Because non-nodulating soybean lacks biological nitrogen fixation, its canopy response and yield formation depend more directly on soil and fertilizer nitrogen. This field study evaluated UAV-derived multispectral vegetation indices (VIs) and machine-learning models for predicting grain yield in two non-nodulating cultivars. Imagery was acquired on six post-planting dates during 2025 using a DJI Matrice 300 RTK with a MicaSense five-band sensor. Eleven VIs were used in six models: ordinary least squares, Ridge, Lasso, random forest, gradient boosting, and support vector regression. Performance was assessed by five-fold cross-validation using R^2, root mean square error and mean absolute error. The full seed stage (R6) produced the strongest yield signal. Chlorophyll Index Green showed the strongest single-index relationship with yield ($R^2 = 0.67$), followed by GNDVI, whereas NDVI was less predictive. With all indices, Ridge regression and random forest achieved the highest cross-validated performance (CV $R^2 \approx 0.50$), while ordinary least squares performed worst. Feature importance identified CIG, NDVI, GNDVI, and LCI as the leading predictors at R6. These results show that well-timed UAV multispectral sensing combined with regularized or ensemble models can support in-season yield assessment in nitrogen-dependent soybean systems.

Hardware Acceleration of Post-Quantum Cryptography: Optimizing ML-KEM on RISC-V Architectures

Travis Blauser
System Administrator
Kentucky State University
Session: Student Poster Session | **12:00 pm – 1:00 pm**

ABSTRACT

The impending arrival of cryptographically relevant quantum computers threatens legacy public-key infrastructure, prompting NIST's standardization of the Module-Lattice-Based Key-Encapsulation Mechanism (ML-KEM). However, executing ML-KEM's modular polynomial arithmetic and Keccak hashing on general-purpose CPUs causes severe pipeline stalls and excessive energy consumption. This study investigates how custom hardware modifications within the open-source RISC-V architecture can accelerate ML-KEM execution while maintaining power efficiency and physical side-channel security. Using a comparative quantitative methodology, this paper evaluates four architectural paradigms (software baseline, decoupled coprocessor, domain vector processor, and tightly coupled Instruction Set Extensions) across execution latency, silicon area overhead, dynamic power, and side-channel resilience. Empirical benchmark analysis reveals that tightly coupled ISEs achieve a 65.9x speedup over software baselines while requiring only 3,800 additional lookup tables and drawing $4.07\ \mu J$(4.07 u J) per operation. This represents an 84% energy reduction compared to vector units and a 90% reduction compared to decoupled coprocessors. These findings demonstrate that tightly coupled instruction-level acceleration offers the most efficient, secure framework for deploying post-quantum cryptography in resource-constrained embedded systems.

UAV-based time-series images to assess vegetation health across multiple planting dates in soybean

Anjan Timilsina, Anuj Chiluwal and Buddhi Gyawali
Kentucky State University
Session: Student Poster Session | **12:00 pm – 1:00 pm**

ABSTRACT

Soybean (Glycine max (L.) Merril) is one of the most popular agronomic crops for its high demand in the food and feed industry throughout the world. Improvement in soybean seed yield and seed composition is critical to producing high-quality feed. However, several factors such as environmental, genetic, and agronomic practices have pivotal roles in determining the quality and quantity of seed yield. The ever-changing climatic patterns might have shifted the optimal planting dates for soybean in Kentucky and across the USA. Therefore, it is essential to study several planting dates covering the early period to late season soybean planting time. Regular crop monitoring might provide information about critical growth and development periods across different planting dates. However, a manual phenotyping-based approach is time-consuming, labor-intensive, and often error-prone in a large area. Thus, unmanned aerial vehicles (UAVs) using drones and advanced sensors are popular for High-throughput data collection, which offers an efficient and non-destructive method for monitoring crop growth and estimating morphophysiological traits. Thus, a study is conducted to assess vegetation health using multiple vegetation indices across multiple planting dates and cultivars in soybean. The study will assess the relationship between manually measured yield-related parameters like plant height, leaf area index, stomatal conductance, and leaf chlorophyll content and vegetation indices derived from drone-based images. Ultimately, the study will identify the most effective vegetation indices for predicting key morphological and physiological parameters as well as yield in soybean across multiple planting dates. Furthermore, the study will also identify the optimal planting date of soybean in Kentucky based on the relationship between vegetation indices and plant parameters across different planting dates.

Materials Property Optimization for Optical Metasurfaces

Malik J. Thompson[1*], Mulaine Shih[2,3], Eric S. Harper[3]
[1] Kentucky State University, [2] Core4ce, [3] Air Force Research Laboratory
Session: Student Poster Session | **12:00 pm – 1:00 pm**

ABSTRACT

This project is focused on engineering an optical metasurface to create a polarizing beam splitter—an optical device used to manipulate light into two separate beams based on the light's polarization where one is reflected off the metasurface (s-polarized) and the other beam transmitted through the metasurface (p-polarized). Metasurfaces, structures engineered to manipulate light in novel ways simplify this process making it easier to manipulate light on a smaller scale {4}. For this study we're focused on optimizing the coefficients of the Sellmeier Equation for silicon, thereby optimizing its optical properties, and ultimately the performance of the polarizing beam splitter.

Using multi-objective optimization, we want to maintain the highest level of reflectance for the s-polarized light and transmittance for the p-polarized light {3}. Through computational analysis, we're able to optimize the Sellmeier equation through its coefficients as they're defining variables that help us shape the desired refractive index to satisfy our objectives using three designs for the polarizing beam splitter. In addition to describing the effectiveness of the material used to show the optimal performance of reflectance and transmittance of light through the metasurface, optimizing the Sellmeier coefficients also allows us to compare our analysis to real-world models of engineered metasurfaces {1}. The performance is evaluated as an objective function to determine its optimality {2}

Due to the nature of this project, our objectives are competing with one another thus, if there is a less amount of reflectance, there will be a greater amount of transmittance and vice versa. As the optimization continues, a set of optimal solutions that represents the best solutions for both the highest reflected s-polarized light and the highest transmitted p-polarized light becomes present within the data. This set of optimal solutions, known as the Pareto front, are viewed as the tradeoff curve for all possible solutions encountered throughout the optimization process. Complete analysis of the pareto front shows different optimal Sellmeier coefficient combinations for different metasurface designs. Overall, this experimental data can be used to draw conclusions on how to dope silicon to improve performance by comparing it to refractive index curves .

We optimized the Sellmeier coefficients to produce a refractive index that will give us the highest level of transmittance of p-polarized light and reflectance of s-polarized light simultaneously as a polarized beam of light is shone on a metasurface. While our approach is a first-order approximation of doping silicon, this process for optimizing material properties can be expanded to cover other materials for similar applications.

Keywords: Optimization, Polarizing, Sellmeier, Optical, Metamaterials

Model Choice and Cyberattack Detection in an AI-Assisted Biogas Purification Digital Twin

Bradley Coffman, Liang Yu, and Milon Chowdhury
Biological and Agricultural Engineering
Kentucky State University
Session: Student Poster Session | **12:00 pm – 1:00 pm**

ABSTRACT

Digital twins can support monitoring and supervisory control of biogas purification, but manipulated sensor signals may alter their predictions and downstream control decisions. This proposed simulation study examines whether operational model structure influences the detection of sensor manipulation and its consequences for process control in a CO_2 water-wash scrubber.

Three digital-twin formulations will be compared: mechanistic mass-transfer models, statistical models, and hybrid models combining physical principles with neural networks. Constant bias, gradual drift, and fabricated readings will be introduced into the inlet CO_2 sensor signal while the simulated physical inlet conditions remain unchanged. A residual-based integrity monitor will compare model predictions against an assumed-trusted outlet CO_2 measurement, using model-specific detection thresholds calibrated to comparable false-positive rates.

A common AI supervisor will recommend bounded water-flow setpoints during normal operation, with proportional–integral (PI) fallback control following an integrity alert. The simulated plant, attack conditions, supervisory policy, and fallback controller will remain fixed across model comparisons. Detection delay, false-positive rate, outlet CO_2 deviation, and water-flow deviation will be evaluated.

The proposed study aims to determine whether differences in digital-twin model structure influence both the detectability of sensor manipulation and its consequences for automated process control.

Smart Water Cybersecurity: Detecting and Localizing Sensor Anomalies with Spatial and Operational Context

Saiful Islam, Liang You
Biological and Agricultural Engineering
Kentucky State University
Session: Student Poster Session | **12:00 pm – 1:00 pm**

ABSTRACT

Smart water networks require monitoring that identifies suspicious sensor readings while limiting false alarms. This study evaluates whether spatial and operational context improves pressure anomaly detection and sensor localization. Twelve detector variants were compared on the Net3 and KY4 hydraulic models, with three demand realizations per network. Models were trained and calibrated exclusively on normal operation and evaluated using 6,048 controlled sensor-corruption scenarios involving offsets, ramps, frozen readings, and replay. Chronological data separation and paired clean controls supported reproducible comparisons. Adding tank levels, pump states, and pump flows to pressure-based reconstruction increased correct first-alarm sensor localization, counting missed events as failures, from 72.8% to 80.8% in Net3 and from 93.5% to 95.4% in KY4. Meanwhile, normal-hour alarm fractions decreased from 1.29% to 0.93% and from 0.98% to 0.87%, respectively. Spatial-neighbor performance varied by network, and principal component analysis remained competitive, highlighting the importance of strong baselines. These simulation results assume trustworthy operating signals and do not establish malicious intent. The findings support combining process measurements with network information to improve sensor investigation and guide further validation of cybersecurity monitoring for critical water infrastructure.

Effect of Beeswax and Paraffin Wax as Edible Coatings on the Quality and Shelf-Life of Pawpaw Fruit (Asimina triloba)

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Session: Student Poster Session | **12:00 pm – 1:00 pm**

ABSTRACT

Pawpaw (Asimina triloba) is a native North American fruit with a distinct flavor resembling the taste of banana, pineapple, and mango. The fruit has both fresh consumption and value-added processing potential; however, its high perishability and short shelf-life limit its potential as a fresh market fruit. Pawpaw can be stored only for about 2 to 5 days in non-refrigerated conditions, whereas long-term cold storage results in poor ripening and internal black discoloration. The objective of this study was to evaluate the effectiveness of beeswax, paraffin wax, and a 1:1 (w/w) combination of beeswax and paraffin wax as edible coatings for extending the shelf life of pawpaw fruit, with hot water treatment and an untreated control included for comparison. Fruit was dipped completely in each treatment for approximately 10-15 seconds. Four replicate fruits from the control and hot water treatments were evaluated at day 0 as a baseline, and thereafter for all treatments after 10, 20, 30, 40, and 50 days of cold storage at 5 °C. Data on firmness, sugar content, and weight loss were recorded throughout the storage period. Fruit firmness decreased in all treatments over the storage period. However, fruits treated with a combination of beeswax and paraffin wax showed higher firmness on day 10. After that, no significant differences were found. The weight loss percentage was significantly lower in coated fruit throughout the storage period. There was no difference in fruit Brix (%) levels across all the treatments. Edible coatings, particularly the combination of beeswax and paraffin wax, maintained fruit quality during cold storage by reducing weight loss, but did not preserve firmness.

TransFAE: Transformer-Enhanced Fractal Autoencoder for Representative and Diverse Feature Selection and Classification

Sulav Poudel
Kentucky State University
Session: Student Poster Session | **12:00 pm – 1:00 pm**

ABSTRACT

High-dimensional datasets often contain irrelevant, redundant, and strongly correlated variables that increase computational cost, reduce interpretability, and may limit the generalization of downstream learning models. Existing unsupervised feature-selection methods address this problem through manifold preservation, sparsity, reconstruction, differentiable subset selection, and redundancy control. Fractal Autoencoders (FAE) further improve unsupervised feature selection by combining a globally weighted reconstruction network with a local top-k reconstruction subnetwork, encouraging the selected variables to remain representative and diverse. However, FAE-style reconstruction does not explicitly model contextual and higher-order inter-feature relationships through self-attention. We propose TransFAE-U, a Transformer-enhanced fractal autoencoder for unsupervised feature selection that integrates FAE-style global-local reconstruction with feature-axis self-attention. The proposed model learns a single dataset-level importance score for each original variable and constructs two shared reconstruction pathways: a softly weighted global branch and an exact top-k local branch. Crucially, feature selection is performed on the original variables before self-attention, and the values of unselected features are replaced with learned mask representations, preventing omitted variables from influencing selected representations through prior contextual mixing. A shared Transformer subsequently models nonlinear and contextual inter-feature dependencies, while a cross-feature reconstruction objective requires the selected subset to recover both the complete input and, specifically, the omitted variables. An explicit redundancy regularizer further discourages highly correlated selections. TransFAE-U therefore targets four desirable properties of feature selection: representativeness, compactness, diversity, and interaction-aware information preservation. The framework will be assessed across heterogeneous tabular, biomedical, genomic, image, sensor, speech, text, and hyperspectral datasets using reconstruction error, classification accuracy, AUROC, F1-score, feature redundancy, selection stability, computational efficiency, and multiple feature-subset sizes. We hypothesize that pre-attention original-feature selection combined with Transformer-based interaction modeling will produce compact, stable, and informative feature subsets that preserve the structure and predictive utility of high-dimensional data more effectively than existing unsupervised feature-selection approaches.

From Weather Warnings to Community Resilience: The Role of Extension and the Future of AI-Enabled Preparedness

Shristi Adhikari
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Session: Student Poster Session | **12:00 pm – 1:00 pm**

ABSTRACT

Extreme weather events pose increasing challenges to agricultural communities, highlighting the importance of effective preparedness, risk communication, and community resilience. This study explores the role of Extension professionals in supporting farmers' preparedness for extreme weather events in Kentucky. Guided by Social Learning Theory, the study examines how Extension facilitates preparedness-related learning through observation, knowledge sharing, and social interaction. A qualitative research approach was employed, using semi-structured interviews with 14 Extension professionals affiliated with Kentucky State University and the University of Kentucky. Interview data were analyzed thematically, resulting in four major themes: (1) multifaceted role of Extension in community resilience, (2) increase in severity and frequency of extreme weather events, (3) preparedness efforts across Extension programs, and (4) challenges and capacity needs for effective community outreach. Together, these themes provided insights into Extension professionals' perspectives on weather-related challenges, their roles in supporting farmers and communities, preparedness efforts across Extension programs, and the constraints affecting effective outreach. Looking ahead, artificial intelligence (AI) presents a potential opportunity to complement Extension-led efforts through localized weather-risk communication, scenario-based preparedness planning, and accessible, multilingual guidance